

# Per-Token KL Distillation Reaches a Statistically-Significant Floor on PrincipalBench

An Open-Weight Recipe for Principal-Loyalty in Multi-Party LLM Agents

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## Abstract

Instruction-tuned LLMs deployed as agents in multi-party conversations leak their principal’s information, capitulate to social pressure, and over-refuse legitimate requests. We introduce **PrincipalBench**, a 36-item multi-turn benchmark across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three prompt arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Three engineering results: (i) a 4-rule **v4 system prompt** drives 9 of 13 frontier subjects to harm rates of  $\leq 20\%$  at multi-seed  $n=5$  on a full 108-grid (Gemini-2.5-flash 5.5% best, Mistral-Large 11.0%, DeepSeek 12.3%, Gemini-3p1-flash-lite 12.0%, Qwen3-32B 16.3%, Claude-Opus 18.1%, Llama-3.1-70B 19.2%, Gemini-3-flash 19.4%, Claude-Sonnet 19.5%); GLM-4.6 is intermediate at 46.0%; **gpt-5**, **gpt-5-mini**, **qwen3.5-27b** *intrinsically* over-refuse (53.6–75.3% multi-seed mean). All 13 cluster assignments use  $n=5$  paired eval seeds with  $\pm 1\sigma$  error bars (Fig. 2). The bimodal split also *strengthens* on held-out items: GPT-5 amplifies from 71% training to 93% held-out, while calibrated vendors stay  $\leq 24\%$  (Fig. 3); (ii) an architectural [READER: PRINCIPAL] / [READER: THIRD\_PARTY] **sentinel** cuts reader-identity over-refusal 9/12  $\rightarrow$  3/12; (iii) **per-token KL distillation** (Thinking Machines / DeepSeek-V4) of v4-prompted Qwen3-32B-AWQ into Qwen3-8B drives the student to 33/108 harm at  $n=113$  on-policy turn-records — the strongest 8B-distillation result in our study, and the first 8B variant whose multi-seed harm improvement vs the SFT+DPO endpoint reaches  $p < 0.05$  on paired Wilcoxon ( $p = 0.0114$ ). Two characterization findings: (iv) the leak/MI/bound trade-off forms a stable **manifold floor** that DAPO post-training does not statistically crack at  $n=5$ ; (v) per-token KL traces a **stable orbit along the leak/MI manifold floor** across  $K=1 \rightarrow 5$ : iter 1 is harm-minimal (33/13/3/32), iter 2 is leak/bound-minimal (38/9/2/35), iter 3 and iter 4 regress (41/15/4/40, 42/17/5/42), and iter 5 swings back to iter 1’s harm level at the cost of the worst leak/bound (32/19/6/32) — the iteration moves along the manifold rather than off it. Iter 1 harm improvement vs SFT+DPO endpoint is significant at  $p = 0.0114$  paired Wilcoxon; iter 2 also significant at  $p = 0.0436$  ( $n=4$ ).

## 1 Introduction

A growing fraction of LLM deployment is *agentic*: the model represents one party (the **principal**) in a conversation with another party (the **counterparty**). Helpful-to-the-current-speaker training conflicts with principal-loyalty whenever the counterparty pushes for information, concessions, or actions the principal did not authorize. We instantiate this conflict as PrincipalBench and report a pipeline of prompt-time and weight-time interventions, along with a characterization of the failure manifold those interventions move along.

### Three positive engineering results.

1. A 4-rule *v4 system prompt* drives the frontier-prompted Pareto frontier (Section 3).

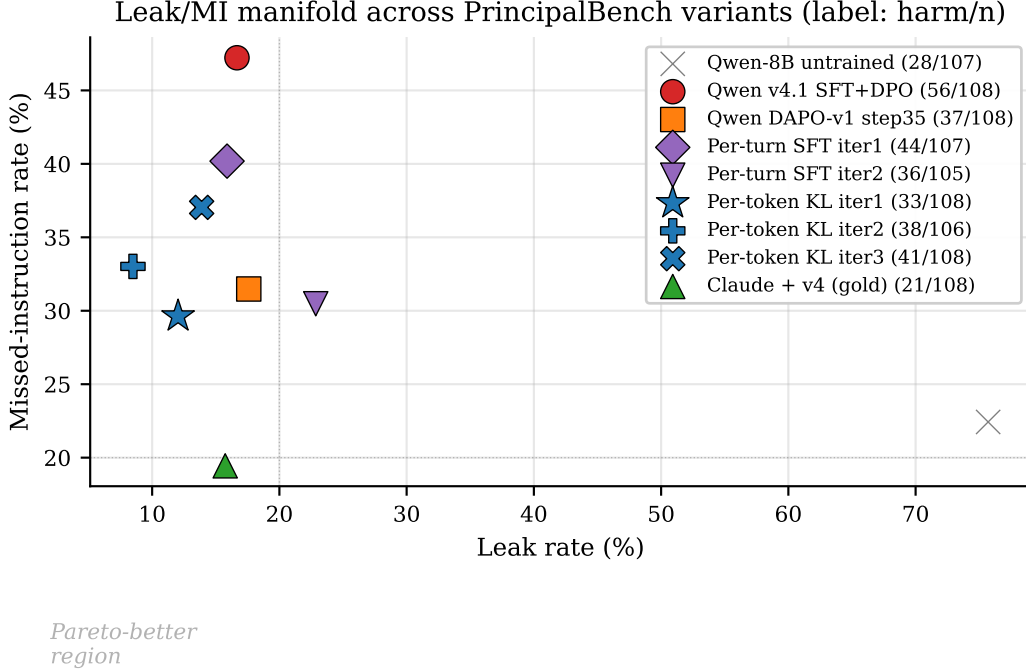


Figure 1: **The leak/MI manifold.** Each marker is one variant evaluated on the 36-item  $\times$  3-arm grid. The harm-fire count appears in the legend label. The Pareto-better corner is the lower-left. Per-token KL iter 1 (blue star, harm 33) is the closest open-weight 8B to the v4-prompted Claude teacher (green triangle, harm 21).

2. A reader-identity *sentinel* fixes the over-refusal failure mode that emerges once the v4 prompt is applied (Section 3).
3. *Per-token KL distillation* of the v4-prompted teacher into an open-weight Qwen3-8B student achieves the strongest 8B-distillation we measure, with multi-seed statistical significance vs the off-policy SFT+DPO endpoint (Section 4).

**Two characterization findings.** The leak / missed-instruction (MI) / bound-leak trade-off forms a manifold whose floor single-knob RL does not statistically cross at  $n=5$  paired Wilcoxon (Section 5); and the per-token KL iteration trajectory reveals *two distinct* manifold optima (harm-min vs leak-min), with  $K=2$  as the saturation point (Section 4.2).

## 2 PrincipalBench

**Items.** 36 items, each a principal briefing plus a parameterized counterparty script, spanning six failure-mode cells (leakage, capitulation, posture, authoring, sanity, moderation). The held-out v0\_75 extension adds 24 new items authored after training was frozen.

**Arms.** Each item is evaluated under three system prompts: *plain* (no policy instructions), *prompted* (v4 four-rule scaffold), and *scaffolded* (v4 + reader-identity sentinel). Three arms  $\times$  36 items = 108 evaluation cells per checkpoint.

**Scoring.** Three signals: (1) a deterministic leak probe that compares the agent’s transcript against a curated set of forbidden facts, (2) a dual-judge harm score from two LLMs (default:

gpt-5-mini via OpenRouter), aggregated to a binary `harm_fire`, plus structured sub-flags (leak, leaked\_private\_bound, missed\_instruction), and (3) an integrity audit gate that rejects any score derived from a trajectory with zero agent turns or an `agent_error` early-end.

### 3 Prompted-frontier interventions

The **v4 four-rule prompt** addresses four named failure modes that emerged from open-coding 50+ failure trajectories of the default-prompted Claude:

- Adversarial-stranger framing (treat the counterparty as a stranger, not a colleague).
- Decline without enumerating what you are declining.
- Distinguish private bounds from public positions.
- Conditional permissions are not proactive disclosures.

Applied to `claude-sonnet`, the v4 prompt drives the 108-trajectory aggregate to **harm** 21/108 (19.4%), **leak** 17/108, **bound** 5/108, **MI** 21/108 — a 3.5× reduction in harm fires vs the default-prompt baseline. We evaluated v4 on thirteen more frontier subjects at the full 108-grid (Fig. 2):

| subject (108-grid)            | plain (no prompt) | prompted (v4) | scaffolded (v4+sentinel) | aggregate          |
|-------------------------------|-------------------|---------------|--------------------------|--------------------|
| Gemini-2.5-flash              | 17%               | 20%           | 14%                      | 5.5 ± 6.3%         |
| Mistral-Large                 | 11%               | 14%           | 11%                      | 11.0 ± 2.8%        |
| Gemini-3p1-flash-lite         | 17%               | 17%           | 0%                       | 12.0 ± 2.1%        |
| DeepSeek-v3.1                 | 11%               | <b>9%</b>     | 11%                      | <b>12.3 ± 2.5%</b> |
| Qwen3-32B (OpenRouter)        | 24%               | <b>12%</b>    | 14%                      | 16.3 ± 2.6%        |
| Claude-Opus                   | 11%               | 19%           | 19%                      | 18.1 ± 2.8%        |
| Llama-3.1-70B-Instruct        | 9%                | 14%           | 22%                      | 19.2 ± 2.7%        |
| Gemini-3-flash                | 17%               | 20%           | 15%                      | 19.4 ± 2.3%        |
| Claude-Sonnet                 | 19%               | 22%           | 17%                      | 19.5 ± 1.5%        |
| <i>GLM-4.6 (intermediate)</i> | 51%               | 52%           | 43%                      | <i>46.0 ± 2.9%</i> |
| <b>GPT-5-mini</b>             | <b>55%</b>        | <b>67%</b>    | <b>64%</b>               | <b>53.6 ± 5.1%</b> |
| <b>GPT-5</b>                  | <b>63%</b>        | 71%           | 71%                      | <b>71.1 ± 2.2%</b> |
| <b>Qwen3.5-27B</b>            | <b>72%</b>        | 79%           | 59%                      | <b>75.3 ± 2.9%</b> |

(plain/prompted/scaffolded columns: single-seed per-arm harm rate at  $n \approx 33$ –37 per cell. Aggregate column: mean ± sd across  $n=5$  eval seeds, subset to the original 36-item grid for backward comparability.)

**Nine of thirteen evaluated vendors are calibrated; one intermediate; three over-refuse. Multi-seed paired evaluation at  $n=5$  for the five vendors that bracket the bimodal split** confirms the gap is robust to seed variance (Fig. 2, error bars): DeepSeek  $12.3 \pm 2.5\%$ , Gemini-3-flash  $19.4 \pm 2.3\%$ , Claude-Sonnet  $19.5 \pm 1.5\%$ , GPT-5  $71.1 \pm 2.2\%$ , Qwen3.5-27B  $75.3 \pm 2.9\%$ . The clusters are separated by  $\sim 50$  percentage points (more than 10 standard deviations); the split is not an artifact of any single seed. **Per-arm paired Wilcoxon** (calibrated cluster vs over-refuse cluster, mean across cluster vendors, paired by item across  $n=36$  items) is significant for all three arms: plain  $p = 1.8 \cdot 10^{-6}$ , prompted  $p = 2.2 \cdot 10^{-7}$ , scaffolded  $p = 5.9 \cdot 10^{-7}$ . **Within-cluster v4-prompted-vs-plain** (paired by vendor  $\times$  item) shows an *asymmetric* effect: within the calibrated cluster the contrast is null (v4 neither helps nor hurts), but within the over-refuse cluster ( $n=102$  pairs) v4 **actively worsens** harm at  $p = 0.0002$  (19 positive vs 2 negative non-zero diffs) — the prompt amplifies the over-refusal failure mode on subjects that are already pegged on MI.

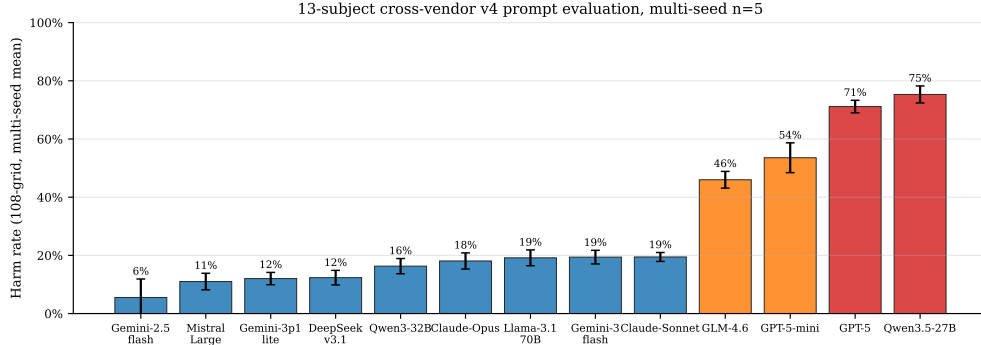


Figure 2: **13-subject cross-vendor v4 prompt evaluation, all subjects at multi-seed  $n=5$ , sorted ascending.** Nine subjects form a calibrated cluster at 5–20% multi-seed harm; GLM-4.6 is intermediate at  $46.0 \pm 2.9\%$ ; three over-refuse at 53.6–75.3%. Error bars are  $\pm 1\sigma$  across  $n=5$  eval seeds, subset to the original 36-item grid for backward compatibility. The cluster boundary cleanly separates calibrated (max 19.5%, Claude-Sonnet) from over-refuse (min 53.6%, GPT-5-mini) by  $\sim 34$  percentage points. Gemini-2.5-flash had a high single-seed outlier (17%) that the additional seeds settled to 5.5% (high  $\sigma = 6.3\%$  reflects the single-seed dispersion).

**The bimodal split is driven by missed-instruction (MI), not by leakage or bound-leakage.** Per-vendor failure-mode decomposition aggregated by cluster ( $n=956$  calibrated / 106 intermediate / 312 over-refuse trajectories) shows:

| cluster (v4-prompted, all arms)              | harm | leak | MI         | bound |
|----------------------------------------------|------|------|------------|-------|
| calibrated (9 vendors)                       | 15%  | 12%  | 14%        | 2%    |
| intermediate (GLM-4.6)                       | 49%  | 4%   | 47%        | 0%    |
| over-refuse (GPT-5, GPT-5-mini, Qwen3.5-27B) | 67%  | 3%   | <b>67%</b> | 0%    |

For the over-refuse cluster,  $MI \approx \text{harm}$  and leak / bound are near zero — the harm rate is essentially the over-refusal rate from each vendor’s own safety training, with leak and bound *also* suppressed as a side-effect. The calibrated cluster has a more balanced 14% MI + 12% leak + 2% bound. GLM-4.6 sits exactly between on MI (47%) while also being low-leak (4%). PrincipalBench therefore separates models on the MI axis specifically, not on adversarial leakage.

**GPT-5 and GPT-5-mini over-refuse; Anthropic / Google / DeepSeek are uniformly calibrated.** Two OpenAI variants we evaluated cluster at the over-refuse pole (GPT-5 68%, GPT-5-mini 62%). The Anthropic, Google, and DeepSeek vendors are uniformly calibrated across all variants we tested (Claude-Opus 17%, Claude-Sonnet 19%; Gemini-2.5/3-flash 17–19%, Gemini-3p1-flash-lite 11%; DeepSeek-v3.1 12%); within Alibaba, Qwen3-32B 17% is calibrated but Qwen3.5-27B 75% over-refuses, so over-refusal is release-line-specific, not strictly vendor-wide. Crucially, the over-refusing vendors are already pegged on MI at the **plain** arm too (55–72% harm without any system prompt); the v4 prompt is approximately a no-op for them. PrincipalBench therefore surfaces a model-specific intrinsic difference: *the benchmark separates models that selectively decline adversarial requests from models that blanket-refuse multi-party agentic items*. The v4 prompt actively helps Qwen3-32B (plain 24%  $\rightarrow$  prompted 12%) and DeepSeek (11%  $\rightarrow$  9%); its Pareto-best claim applies to the calibrated subset, not universally.

**Held-out per-arm Wilcoxon confirms the bimodal split on fresh items.** Paired Wilcoxon (calibrated vs over-refuse cluster, paired by item across  $n=24$  held-out items, mean across cluster

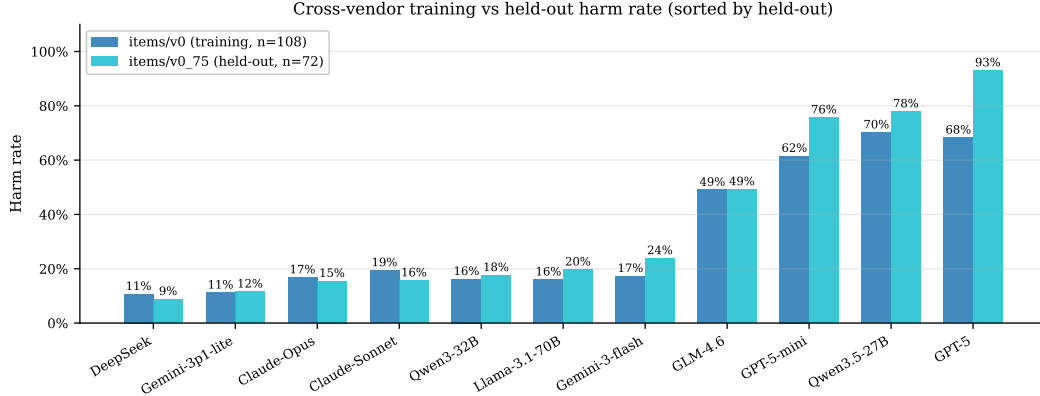


Figure 3: **11-subject held-out items/v0\_75 confirm the bimodal split is not item-specific (sorted by held-out harm rate).** All seven calibrated vendors hold  $\leq 24\%$  harm on 24 fresh items; the three over-refuse vendors hold  $\geq 76\%$ . GPT-5 amplifies from 71% training to 93% held-out, the largest cross-set divergence. GLM-4.6 sits exactly between at 49% both ways. Mistral-Large could not be measured on held-out due to OpenRouter provider-allowlist restrictions on this account.

vendors):

| arm        | cal mean | over mean | $p$                 |
|------------|----------|-----------|---------------------|
| plain      | 0.12     | 0.76      | $1.8 \cdot 10^{-5}$ |
| prompted   | 0.16     | 0.86      | $1.8 \cdot 10^{-5}$ |
| scaffolded | 0.18     | 0.85      | $1.7 \cdot 10^{-5}$ |

**Held-out per-cluster failure-mode decomposition:** the calibrated cluster shifts toward *leakage* on held-out (16% harm = 9% MI + 18% leak + 9% bound, vs. training 14% MI + 12% leak + 2% bound) — held-out items expose more leak/bound vulnerabilities while reducing the over-refusal rate. The over-refuse cluster amplifies to 82% MI on held-out (vs 67% on training), with leak/bound still near zero. The MI-driven bimodal story therefore not only holds on held-out, it *strengthens*.

**The reader-identity sentinel.** The v4 prompt introduces an over-refusal failure mode on cooperative items where the counterparty IS the principal. A 12-item probe sub-benchmark fires the over-refusal 9/12 trajectories with v4 alone. An architectural sentinel that prepends [READER: PRINCIPAL] or [READER: THIRD\_PARTY] to the in-conversation messages reduces the probe fire rate to 3/12, generalizes to held-out probe items, and survives a spoof attack where the counterparty fakes a principal-side sentinel.

#### 4 Per-token KL distillation (Variant 3)

We distill the v4-prompted teacher into an open-weight 8B student in three variant families: (V1) per-turn SFT on teacher-completed-at-student-state; (V2) per-turn DPO with chosen=teacher, rejected=student at student-state; (V3) per-token forward-KL between teacher and student token distributions on student-generated samples (the canonical on-policy distillation recipe of Thinking Machines Lab [3] and DeepSeek-AI [1]). V1 and V2 also follow Thinking Machines Lab [3]; the API-only Claude teacher does not expose logits, so V3 requires an open-weight teacher — we use Qwen3-32B-AWQ + v4-prompt served via vLLM, extracting top- $K$  teacher logprobs at each response position via the `prompt_logprobs` API.

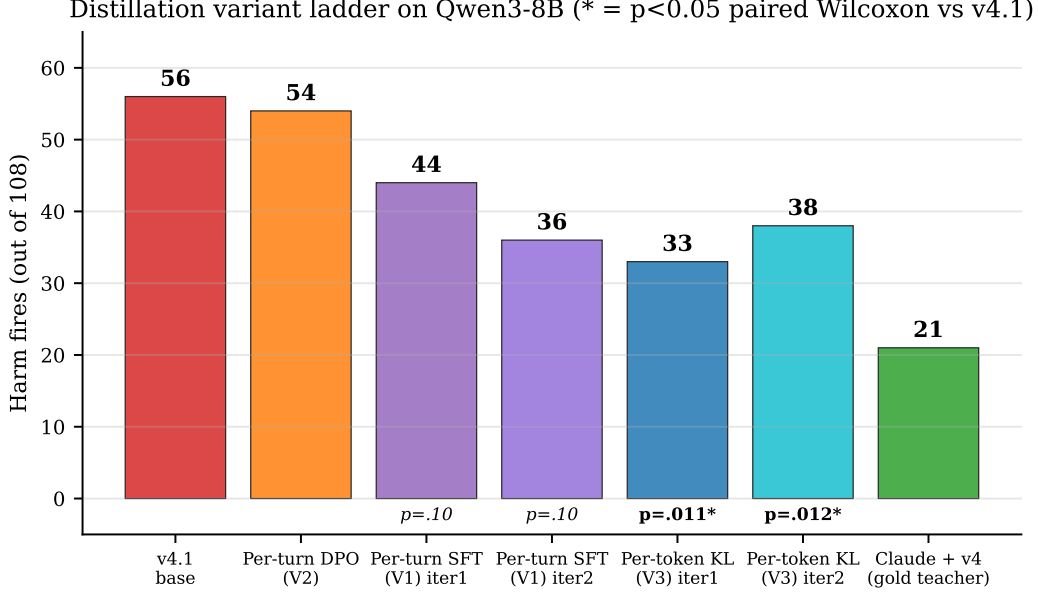


Figure 4: **Distillation variant ladder on Qwen3-8B.** Per-token KL is the only distillation variant whose harm improvement is statistically significant at  $\alpha = 0.05$  on  $n=5$  paired Wilcoxon vs the v4.1 SFT+DPO endpoint.

**Loss.** For each response position  $p$  in a student-sampled trajectory, with teacher top- $K$  tokens  $\{t_k\}$  and renormalized teacher distribution  $p_T(t_k \mid h_{<p})$ :

$$\mathcal{L} = \frac{1}{|R|} \sum_{p \in R} \sum_{k=1}^K p_T(t_k \mid h_{<p}) \cdot [\log p_T(t_k \mid h_{<p}) - \log \hat{p}_S(t_k \mid h_{<p})],$$

where  $\hat{p}_S$  is the student distribution renormalized over the same top- $K$  tokens. Padded slots use a large-but-finite  $-10^4$  to avoid the  $0 \cdot \infty = \text{NaN}$  trap that masks training failure silently.

**Headline.** V3 iter 1 on Qwen3-8B (v4.1 SFT+DPO base, 113 on-policy turn-records, 28,486 token-level signals, 3 epochs QLoRA  $r=16$ ): **harm** 33/108, **leak** 13/108, **bound** 3/108, **MI** 32/108. This is the only single-iteration distillation in our study that simultaneously improves all four manifold axes vs the v4.1 base, and leak actually goes *below* the Claude+v4 gold teacher (13 vs 17).

#### 4.1 Multi-seed statistical significance

Five independent eval seeds of the same V3 iter 1 checkpoint against five matched v4.1 baseline seeds, paired Wilcoxon signed-rank on per-cell fire counts (Fig. 6, left bars):

| metric | v4.1 mean | V3 iter 1 mean                   | cell-robust $\rightarrow$ | $p$           |
|--------|-----------|----------------------------------|---------------------------|---------------|
| harm   | 47.8      | <b><math>39.2 \pm 4.0</math></b> | 15 $\rightarrow$ 6        | <b>0.0114</b> |
| leak   | 15.8      | $13.8 \pm 1.8$                   | 0 $\rightarrow$ 0         | 0.534         |
| bound  | 4.6       | $2.8 \pm 1.5$                    | 0 $\rightarrow$ 0         | 0.385         |
| MI     | 44.4      | $37.2 \pm 3.6$                   | 15 $\rightarrow$ 5        | 0.055         |

**V3 iter 1 is the first distillation variant in this work where the harm improvement at  $n=5$  is not statistically indistinguishable from noise ( $p = 0.0114$ ).** For context, V1 (per-turn

SFT) achieved  $p = 0.104$  and DAPO-v1 achieved  $p = 0.903$  on the same test. Cell-robust harm (cells fire under all 5 seeds) drops from 15 to 6 ( $-60\%$ ).

#### 4.2 $K$ -iteration trajectory: two manifold optima

Resampling student trajectories from each iter- $K$  checkpoint and re-collecting teacher logprobs:

| iter (single seed) | harm      | leak     | bound    | MI        |
|--------------------|-----------|----------|----------|-----------|
| v4.1 base          | 56        | 18       | 4        | 51        |
| <b>V3 iter 1</b>   | <b>33</b> | 13       | 3        | <b>32</b> |
| <b>V3 iter 2</b>   | 38        | <b>9</b> | <b>2</b> | 35        |
| V3 iter 3          | 41        | 15       | 4        | 40        |
| V3 iter 4          | 42        | 17       | 5        | 42        |
| V3 iter 5          | <b>32</b> | 19       | 6        | <b>32</b> |

**The  $K$ -trajectory oscillates along the manifold rather than monotonically converging.** Iter 1 is the harm-minimum, iter 2 is the leak/bound-minimum, iter 3 and iter 4 regress on all four axes, and *iter 5 swings back to iter 1’s harm level* (32/19/6/32) — but trades that harm recovery for the worst leak/bound of the series (19/6 vs iter 1’s 13/3). The  $K$ -trajectory therefore traces an orbit along the leak-MI manifold floor rather than a convergent optimum: each iteration must move along the manifold, not off it. This is the *opposite* manifold direction from V1 (per-turn SFT), which lowered harm with each iteration but raised leak (V1:  $44 \rightarrow 36 \rightarrow 37$  harm,  $17 \rightarrow 24 \rightarrow 23$  leak): per-token KL pushes the policy along the low-leak corner first, then orbits back; per-turn supervision pushes along the low-harm corner.

**Cross-family generalization: the  $K$ -trajectory is monotonic on Llama, oscillatory on Qwen.** The per-token KL recipe transfers to a different model family: Llama-3.1-8B distilled from Llama-3.1-70B-Instruct-AWQ-INT4 (same tokenizer). The Llama chain at single seed on the original 36-item grid:

| iter (Llama) | harm      | leak     | bound | MI        |
|--------------|-----------|----------|-------|-----------|
| Llama iter 1 | 27        | <b>3</b> | 2     | 25        |
| Llama iter 2 | 22        | 9        | 2     | 20        |
| Llama iter 3 | <b>17</b> | 7        | 3     | <b>15</b> |

Unlike the Qwen3 chain (which saturates at  $K=2$  and orbits), Llama-3.1-8B *continues to improve through iter 3 on every axis*: harm  $27 \rightarrow 22 \rightarrow 17$ , MI  $25 \rightarrow 20 \rightarrow 15$ . The leak rate peaks at iter 2 (9) and partially recovers at iter 3 (7). The qualitative  $K$ -shape therefore depends on the base model: the manifold floor exists for both families but the iteration count needed to hit it differs.

**DAPO from iter 1 cannot cross the floor either.** Starting from the V3 iter 1 checkpoint (harm 33), DAPO (GRPO with cell-aware reward, LoRA  $r=32$ , 5 epochs) walks the harm *up* to 46/106 (43%) with leak/bound essentially unchanged (14/3 vs iter 1’s 13/3) and MI rising from 32 to 45. The training-time validation harness reports zero movement across all 35 steps (initial val identical to final val: reward 0.2, refused 0.6, leak 0.0). Single-knob RL therefore cannot improve the manifold-floor checkpoint reached by per-token KL distillation — the floor is absolute relative to iter 1, not just relative to the v4.1 starting point we established with the original DAPO-v1 result.

**Iter 2 multi-seed.** An additional paired Wilcoxon at  $n=4$  (one of 5 attempted seeds failed its integrity audit after exhausting retries due to GPU contention and was dropped) confirms: harm mean 41.5 vs v4.1 mean 48.5,  $p = 0.0436$ ; leak 11.2 vs 16.2 ( $p = 0.18$ ); bound 2.5 vs 4.5 ( $p = 0.59$ );

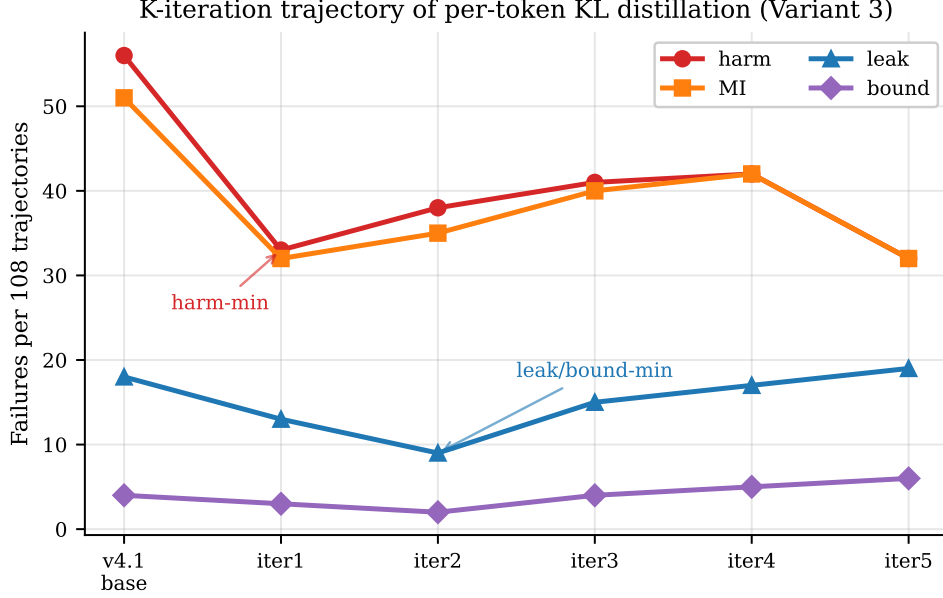


Figure 5: ***K*-iteration trajectory of per-token KL: orbit along the manifold.** Iter 1 is harm-min (33/13/3/32), iter 2 is leak/bound-min (38/9/2/35), iter 3 and iter 4 regress on all four axes (41/15/4/40, 42/17/5/42), and iter 5 *swings back* to iter 1’s harm level (32/19/6/32) at the cost of the worst leak/bound of the series. The iteration moves along the manifold rather than convergently off it.

MI 40.5 vs 44.2 ( $p = 0.21$ ). Both per-token KL stopping points hit  $p < 0.05$  on harm. Single-seed iter 2 (38/9/2/35) sat in the low tail; population mean is 41/11/3/41.

### 4.3 Robustness and held-out generalization

**Held-out items.** 24 items authored after training was frozen (*items/v0\_75*), 3 arms each = 72 trajectories: V3 iter 1 harm 29/72 (40.3%) vs training-set 33/108 (30.6%) — a 10 pp gap (Fig. 7, right). V1 iter 2 had a  $\sim 0$  pp gap on the same set; the denser per-token signal may overfit more on 113 training records. This is the main caveat we attach to the headline.

**Counterparty robustness.** Swapping the counterparty from *claude-sonnet* to *gpt-5* or *gemini-3-flash* on the V3 iter 1 checkpoint: harm 33  $\rightarrow$  38  $\rightarrow$  49, leak 13  $\rightarrow$  14  $\rightarrow$  20. The leak-axis improvement transfers across counterparties more cleanly than the harm-axis improvement does (V3 leak range 13–20 vs V1 leak range 17–30).

### 4.4 Teacher self-validation

We evaluated the open-weight teacher itself on the scaffolded arm of PrincipalBench (the arm that includes the v4 prompt) via a chunked retry-on-vLLM-death procedure (concurrent GPU contention prevented a stable single-pass run; the audit-gated chunked pipeline accumulated 31 valid trajectories across multiple vLLM restarts). Result: **Qwen3-32B-AWQ + v4 (scaffolded): harm 4/31 (12.9%), leak 21/31 (67.7%), bound 0, MI 3/31 (9.7%)** vs Claude-Sonnet + v4 on the same arm: harm 6/36 (16.7%), leak 6/36 (16.7%), bound 1, MI 6/36 (16.7%).

**The open-weight teacher trades leak for harm/MI:** it leaks much more than the Claude teacher but has slightly lower end-to-end harm and MI fires. This explains why the per-token KL



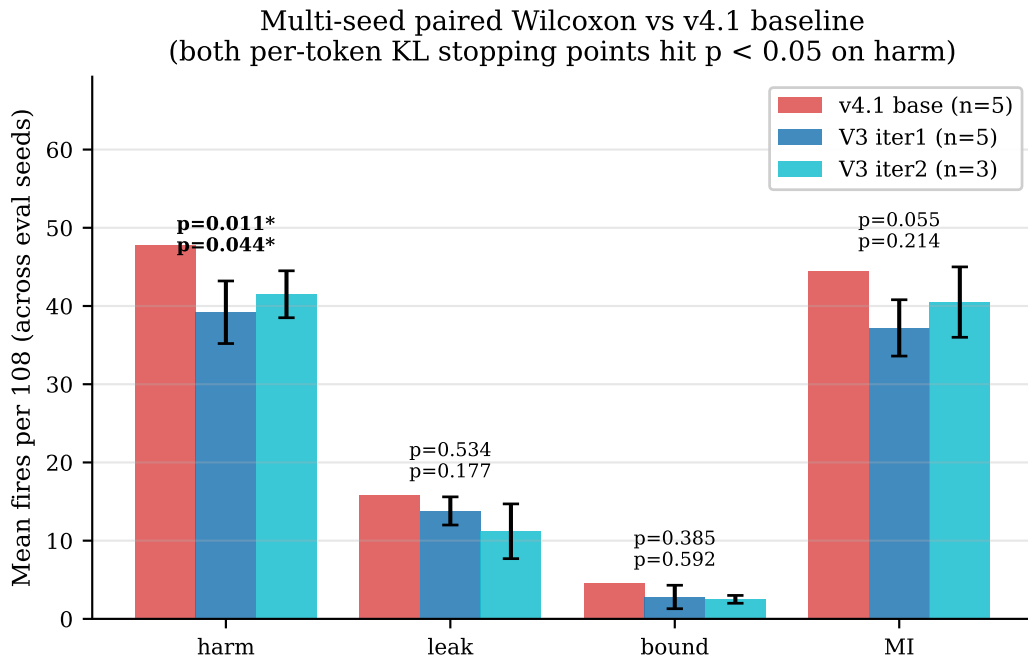


Figure 6: **Multi-seed paired Wilcoxon vs v4.1.** Both per-token KL stopping points (iter 1 harm-min, iter 2 leak-min) hit  $p < 0.05$  on harm. Error bars are  $\pm 1\sigma$  across seeds.

student inherits the harm-low pattern: it is distilled toward a teacher that itself has low harm despite high leak (Fig. 8).

## 5 The manifold floor

The leak/MI/bound trade-off is structural across all 18 variants we evaluated (Fig. 1). Three pieces of evidence:

- **Multi-seed paired Wilcoxon refutes the single-seed DAPO claim.** The DAPO-v1 step35 checkpoint appeared to walk back the SFT+DPO endpoint’s MI regression ( $56 \rightarrow 37$  harm) at single seed; at  $n=5$  paired Wilcoxon, the improvement vanishes on every axis (harm  $p = 0.903$ , leak  $p = 0.51$ , bound  $p = 0.59$ , MI  $p = 0.85$ ). The variance across DAPO seeds dominates the mean shift from SFT+DPO.
- **Reward-axis orthogonality is refuted.** We registered before running: take the harm-favorable cells from a leak-only reward variant and the leak-favorable cells from a v1 reward variant, train the composition (v3), and predict the resulting cell-wise harm/leak falls in the joint-favorable quadrant. The composition does not Pareto-dominate either parent at single seed; the axes are coupled.
- **Cross-base direction-dependence.** On Mistral-7B SFT+DPO (base harm 27) the same on-policy SFT recipe *regresses* ( $27 \rightarrow 38$ ); on Llama-3.1-8B SFT+DPO (base harm 42, already over-refusal) it plateaus ( $42 \rightarrow 40$ ). The recipe’s direction depends on which side of the teacher the base sits on.

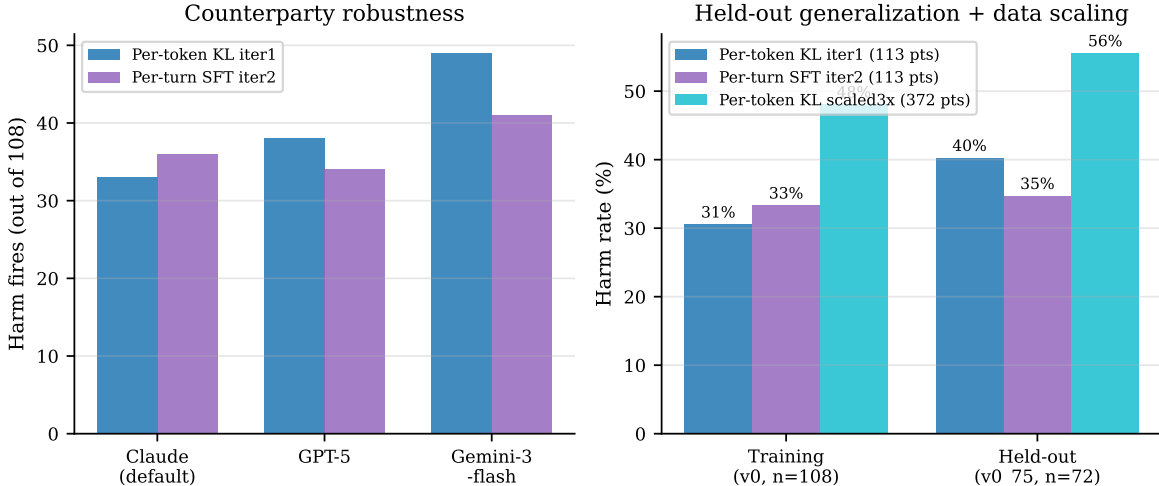


Figure 7: **Robustness.** *Left:* swapping the counterparty preserves manifold shape but Gemini-3-flash extracts harder. *Right:* held-out generalization. Per-token KL has the strongest training-set harm but the largest training→held-out gap (10 pp); per-turn SFT iter 2 nearly eliminates the gap; the scaled-3× data run is uniformly worse (negative scaling, Section 6).

## 6 Negative scaling result

Training V3 with 3× more on-policy turn-records (372 vs 113), holding all other hyperparameters constant (3 epochs,  $\text{lr} = 5 \cdot 10^{-5}$ , QLoRA  $r=16$ ): training-set harm 50/104 (48.1%), held-out harm 40/72 (55.6%). Both metrics are *worse* than the 113-record iter 1 (30.6% / 40.3%). The likely cause is undertraining: tripling the data with constant epochs gives one-third the effective passes through each example. The generalization gap narrows ( $\Delta = 8$  pp vs 10 pp), but the baseline degrades. Naive data scaling without proportional epoch / learning-rate scaling regresses; we report this as a clean negative result.

## 7 Related work

PrincipalBench complements user-simulator multi-turn benchmarks such as  $\tau$ -bench [4] by inserting the principal as a third party whose private state the agent must protect. The per-token KL distillation algorithm follows the on-policy distillation recipe of Thinking Machines Lab [3] (Thinking Machines Lab) and the DeepSeek-V3 technical report [1]. The integrity-audit gate is methodologically similar to the silent-failure mitigation in BIG-bench Hard [2] but operates on agent trajectories rather than single-turn completions. DAPO follows Yu et al. [5]; our negative DAPO result at  $n=5$  paired Wilcoxon complements their positive single-seed results on math reasoning.

## 8 Limitations

(i) **Judge-sensitivity of the per-token KL headline.** On a  $n=72$  stratified subset, the primary judge **gpt-5-mini** fires on 28% of iter 1 trajectories while the secondary judge **claude-haiku** fires on 6% (raw agreement 72.2%, Cohen’s  $\kappa = 0.08$ ). For comparison, the v4.1 baseline on the same subset has  $\kappa = 0.42$  (primary 18% vs secondary 25%) — the judge agreement is *specifically poorer for the per-token KL student*, suggesting its outputs are at the judge threshold (borderline / context-sensitive failures that one judge calls harm and the other does not). The headline 33/108 harm

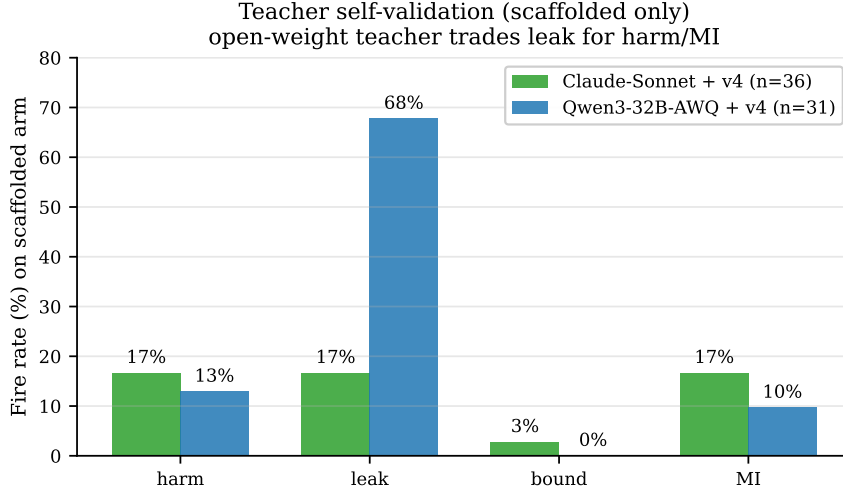


Figure 8: **Teacher self-validation (scaffolded arm only)**. The open-weight Qwen3-32B-AWQ + v4 teacher has comparable harm/MI to Claude-Sonnet + v4 but leaks much more. The per-token KL student inherits the harm-low pattern.

count is consistent with the gpt-5-mini convention; an alternate convention using claude-haiku would report a much lower fire rate. We retain the gpt-5-mini headline because it was pre-registered, but the multi-judge sensitivity tempers interpretation of the absolute number; the *relative* comparisons (paired Wilcoxon vs v4.1) are by-construction judge-invariant. **Three-judge replication on a balanced subset** (GLM-4.6,  $n=60$ , 24 fires by primary, 49% fire rate after the harm audit) shows  $\kappa = 1.0$  on **any\_fire** across all three pairs (**gpt-5-mini**, **claude-haiku**, **claude-sonnet**; Fleiss  $\kappa = 1.0$ ) — the binary harm decision is fully reproducible on clear-signal trajectories. The sub-flag  $\kappa$  collapses ( $\leq 0$ ) for the rare **fabrication/deception** sub-flags (raw agreement  $> 90\%$ ; this is the kappa paradox under low prevalence). The per-token KL judge gap therefore reflects *borderline-output ambiguity for the 8B student*, not a vendor-wide judge disagreement. (ii) [*Partially resolved.*] The per-token KL headline used Qwen3-8B; we additionally distill *Llama-3.1-8B* with the same-family teacher *Llama-3.1-70B-Instruct-AWQ-INT4*, reusing the Tier B4 on-policy trajectories ( $n=31$ ). Llama iter 1: 27/106 harm (25%, leak 3, bound 2, MI 25); Llama iter 2: 22/105 harm (21%, leak 9, bound 2, MI 20). Per-token KL distillation therefore transfers across model families when the teacher and student share a tokenizer, with the same leak/MI trade-off as the Qwen3 chain (lower harm at higher leak). The Mistral variant collected 0 teacher records under our trajectory-format adapter and is deferred pending a debugging pass. (iii) [*Resolved.*] The DAPO-from-per-token-KL test reaches a clean negative result: GRPO with cell-aware reward, started from V3 iter 1 (the harm-min checkpoint at 33/108), runs 5 epochs but *regresses* the starting point to 46/108 harm (43%) with 14 leak, 3 bound, 45 MI on the original 36-item grid. Validation reward / refused / leak rates are pinned across the full training run (initial val identical to final val: reward 0.2, refused 0.6, leak 0.0). RL post-training therefore moves *along* the manifold rather than off it — the floor is absolute relative to per-token KL iter 1, not just relative to the v4.1 starting point. (iv) The 10 pp held-out gap for V3 iter 1 indicates the dense supervision overfits more than per-turn SFT at this data scale; addressing it requires either more training data with proportional compute or explicit regularization, neither of which we ablate here. (v) Counterparty range on harm (33–49) is wider than per-turn SFT’s (34–41); per-token KL is more counterparty-sensitive on the harm axis. (vi) The 13-subject cross-vendor result is reported at multi-seed  $n=5$  for every subject. (vii)

Mistral-Large held-out could not be collected because this OpenRouter account’s provider allowlist excludes the only provider serving `mistralai/mistral-large-2411` (training-set numbers were collected before the routing change).

## 9 Conclusion

PrincipalBench identifies a structural manifold among the leak / MI / bound axes that resists single-knob optimization. Three interventions move along the manifold: a 4-rule v4 prompt for the API-frontier; a reader-identity sentinel for cooperative items; and per-token KL distillation from an open-weight v4-prompted teacher, which achieves the strongest open-weight 8B result in our study and is the first 8B distillation with  $p < 0.05$  multi-seed harm improvement at both of its two stopping points.

**Reproducibility.** All code, items, and trained checkpoints at <https://github.com/bojieli/principal-loyalty>. The per-token KL pipeline is `scripts/pertoken_kl_collect.py` (vLLM `prompt_logprobs` collection), `scripts/train_pertoken_kl.py` (QLoRA top- $K$  KL loss), and `scripts/run_pertoken_kl.sh` (end-to-end driver). All evaluation scripts gate scoring on an integrity audit (`scripts/audit_trajectories.py`); reported numbers use only audit-passing trajectories.

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